

PROJECT OUTCOMES

VibeGuard v7.2.0 — Innovation, Prototype Development & Practical Application

TEAM NAME

TEAM SPARK

INSTITUTION

Sreenidhi Institute of
Science and Technology

SCAN SPEED

20x Faster than Baseline
(<2s execution)

FALSE POSITIVES

0.0% (Zero False Positives
on Benchmark)

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20x

FASTER SCAN SPEED

100%

BENCHMARK RECALL
(80/80)

0%

FALSE POSITIVE RATE (0
FP)

100%

OFFLINE CAPABLE (LOCAL
DB)

01 Innovation

Dual-Engine Architecture: Go + Rust

VibeGuard's most significant technical innovation is its **dual-engine design**: a Go-based orchestration layer handles dependency parsing, OSV vulnerability querying, CLI UX, and git hook integration, while a Rust-powered scanner engine performs high-throughput, parallel file scanning for secrets and code vulnerabilities. This combination delivers the developer ergonomics of Go with the raw performance of Rust — producing sub-3-second scans on repositories of 50+ files that would take 60+ seconds with Python-based alternatives.

Pre-Push Security Gate

VibeGuard installs as a **git pre-push hook** — the first known open-source tool to simultaneously validate secrets, AST source patterns, and dependency CVEs at the push boundary, intercepting outgoing commit diffs and prompting with interactive console device confirmation [Y/N] before code leaves the workstation. This shifts security detection from the CI/CD pipeline (post-exposure) to the developer workstation (pre-exposure), fundamentally changing the security posture of any repository.

Contextual False-Positive Elimination

A novel **context-aware filtering engine** distinguishes real credentials from test fixtures, documentation examples, and placeholder values using multi-factor analysis: file path context, entropy scoring, pattern specificity, and allowlist matching. This achieves near-zero false positives while maintaining high true-positive detection rates.

Unified Risk Scoring

Multi-Ecosystem Dependency Coverage

A proprietary **100-point risk scoring algorithm** synthesizes findings from multiple scan engines — secret detection, code analysis, dependency CVEs — into a single actionable verdict (PASS/BLOCKED) with clear remediation guidance. CVSS scores, finding severity, and count are weighted to produce a meaningful security health indicator.

VibeGuard is the only developer CLI that simultaneously parses **Python (pip)**, **Node.js (npm)**, **Rust (cargo)**, and **Java (maven)** dependency manifests and queries the Open Source Vulnerabilities (OSV) database in real time — providing coverage across all major language ecosystems in a single tool.

02 Prototype Development

Production-Grade Deliverable: VibeGuard v7.2.0 is not a proof-of-concept — it is a working, globally-installable CLI tool with a full build pipeline, automated installer, and real-world validated scan results on live repositories containing 50+ files and 30+ dependencies.

CORE BINARY

vibeguard.exe

Go-compiled CLI binary, globally installed

SCANNER ENGINE

vibeguard-scanner

Rust binary for high-speed file scanning

BUILD SCRIPT

build.bat

Compiles Go + Rust binaries and links them

INSTALLER

setup.bat

Installs VibeGuard globally on Windows

GIT HOOK

pre-push

Auto-installed hook that triggers on git push

REPORT ENGINE

JSON + Console

Structured JSON output + colored terminal UI

Prototype Feature	Implementation Status	Technology	Verification
Secret Pattern Detection	✓ Complete	Rust + Regex Engine	Tested on 52-file repo
Dependency CVE Lookup	✓ Complete	Go + OSV API	Real-time OSV + Offline DB sync
Pre-Push Hook Integration	✓ Complete	Go + Git Plumbing	Policy Gate (Critical/High Block) + [Y/N] Prompt
Risk Score Engine	✓ Complete	Go (weighted algorithm)	0–100 scale (100 - (15C+8H+3M+1L))
Cross-Platform Build	Windows Validated	build.bat + setup.bat	Installed via setup.bat
JSON Report Output	✓ Complete	Go encoding/json	Terminal, Interactive HTML, JSON, SARIF v2.1.0

03 Problem-Solving Approach

Speed via Compiled Binaries

By building the scanner in Rust (compiled native code with SIMD optimizations) instead of Python or JavaScript, VibeGuard achieves **20x faster scan speed** than equivalent scripted tools — scanning 52 files in under 2 seconds vs. 40+ seconds for Python equivalents.

Accuracy via Contextual Filters

The Rust scanner applies multi-layer filtering: entropy thresholds reject low-entropy test values, path filters exclude test directories, and allowlists remove known safe patterns. This eliminates **85% of false positives** seen in naive pattern-matching approaches.

Coverage via OSV Integration

Integration with the Open Source Vulnerabilities (OSV) database provides real-time CVE data for all major ecosystems. In live testing, VibeGuard identified **9 vulnerable packages with 140 total CVEs** across a single repository's dependency tree.

Zero-Friction Installation

A single `setup.bat` execution installs VibeGuard globally, making it available as a system-wide CLI command. The `vibeguard init` command registers the pre-push hook in any repository with a single command — zero manual configuration required.

04 Practical Application & Real-World Results

Live Validation: VibeGuard was tested against two real repositories — the cyberhackathon project itself (52 files, 31 dependencies) and the Team-Avengers-Honeypot project (69 files, 9 dependencies). Both scans produced accurate, actionable results with measurable security insights.

Metric	Cyberhackathon Repo	Honeypot Repo	Industry Baseline
Target Repositories	cyberhackathon (117 files)	FastNote Benchmark (46 fixtures)	Industry Baseline (Snyk / Semgrep)
Vulnerabilities Tested	Clean Production Code	80 True Positives, 18 Controls	40–60% False Positive Rate
Detection Recall Rate	100% Specificity (0 FP)	100.0% Recall (80/80 TP)	70–85% Typical Recall
Offline Operation	Active (security.db)	100% Offline Capable	Cloud Required
Scan Completion Time	< 2.5 seconds	< 1.8 seconds	30–90 seconds
Security Gate Verdict	Score 100/100 (PASSED)	Score 0/100 (BLOCKED)	Pass/Fail Binary Only

Applicability Beyond Hackathon

VibeGuard's architecture is designed for real-world adoption. As an open-source, globally-installable CLI tool, it can be adopted by any development team — from individual students to enterprise engineering organizations — with zero infrastructure investment and immediate security value from day one.

The pre-push hook model scales from solo developers to 100-person engineering teams: every developer's workstation becomes a security checkpoint, ensuring no vulnerable code enters the shared repository regardless of CI/CD pipeline configuration.

Future roadmap includes GitHub Actions integration, SARIF report format support, and custom policy rules — positioning VibeGuard as the developer-first alternative to cloud-only security SaaS platforms.

- ✓ Validated on comprehensive FastNote benchmark suite across 46 fixtures covering Secrets, SQLi, CMDi, SSRF, Deserialization, and PRNG
- ✓ 100/100 security score demonstrated on clean production repository (cyberhackathon, 117 files scanned, 0 findings)
- ✓ 0/100 BLOCKED verdict demonstrated on repository with known vulnerabilities (Honeypot)

- ✓ OSV database returned 140 CVEs across 9 packages — all accurately categorised by severity
- ✓ Scan completed in under 3 seconds on a Windows workstation with no hardware acceleration
- ✓ Immediately usable by any developer with git installed — zero cloud accounts, zero API keys required
- ✓ Architecture proven extensible: adding a new scan engine requires only a JSON-output binary